



Unidad Profesional
Interdisciplinaria en Ingeniería y
Tecnologías Avanzadas

A
upiita

Multimedia

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Practica 3 Cambio de colores



Interpretación

- Código



El resultado de este código nos ayudara a cambiar en tonalidad la imagen de manera que se muestre en tonalidad negro y blanco

The screenshot shows a Jupyter Notebook with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The code editor contains a Python script titled "#Binary filter: Black and White". The script opens two files: 'example001.bmp' for reading and 'example001bin.bmp' for writing. It reads the metadata of the first file and writes it to the second. Then, it defines two color arrays: 'blanco' (white) as [0xff, 0xff, 0xff] and 'negro' (black) as [0x00, 0x00, 0x00]. The script then iterates over the pixel data of the first file, converting each pixel to either white or black based on its value. The terminal at the bottom shows the output: "No Pixels: 256".

```
#Binary filter: Black and White

file = open('./images/example001.bmp', 'rb')
fileo = open('./images/example001bin.bmp', 'wb')
metadata = file.read(54)
fileo.write(metadata)
blanco = [0xff, 0xff, 0xff]
negro = [0x00, 0x00, 0x00]

file.seek(54, 0)
no_pix = 0
limite = (pow(2, 24) - 1) / 2
while(True):
    pixel_data = file.read(3)
    if(len(pixel_data) > 0):
        valor_int = int.from_bytes(bytes(pixel_data), byteorder='little')
        if(valor_int < limite):
            fileo.write(bytes(blanco))
        else:
            fileo.write(bytes(negro))
        no_pix += 1
    else:
        break
print('No Pixels: ' + str(no_pix))
file.close()
fileo.close()
```

La imagen a cambiar es la siguiente:



Este es el resultado obtenido: que igual esta formada por 256

